

Shot Folder Auto-Creator - Level 2

Objective: Upgrade your Level-1 CLI tool so that the folder structure is **NO LONGER hard-coded** inside the script, but is loaded from an external **JSON config file**.

In real VFX pipelines, folder structures are not fixed inside code.

Studios store folder rules in **config files**, so they can change structures without modifying Python scripts.

You will now convert your Level-1 tool into a **config-driven tool** that reads a `shot_structure.json` file and creates folders accordingly.

Requirements

- 1 Script usage stays the same

```
python shotsetup.py <shot_name> <root_path>
```

Example run:

```
python shotsetup.py SH050 D:\Projects\Seq01
```

- 2 The folder structure must come from a JSON file

You must create a file named:

```
shot_structure.json
```

Example content (you may use this or your own):

```
{
  "plate": {},
  "comp": {
    "scripts": {},
    "renders": {}
  },
  "cache": {
    "fx": {},
    "anim": {}
  },
  "publish": {
    "images": {},
    "videos": {}
  }
}
```

- ✓ Your script must **read this JSON file**
- ✓ Your script must create every folder shown in the file
- ✓ Must support **nested folders** (multi-level recursion)
- ✗ You CANNOT hard-code any folder names inside Python
- ✗ If config file changes, the tool MUST adapt automatically

3 Error Handling Rules (Mandatory)

Case	Expected Output
JSON file missing	Error: shot_structure.json not found
JSON file unreadable / invalid	Error: Cannot read config file
Invalid root path	Invalid root path!
Shot already exists	Shot already exists, skipping
Folder created	Created folder: SH050/comp/scripts

- ✓ In Level-2 you only **print messages** (logging comes in Level-3)

Example Output (Terminal)

```
Reading folder structure from shot_structure.json...
Creating shot: SH050
Created folder: SH050/plate
Created folder: SH050/comp
Created folder: SH050/comp/scripts
Created folder: SH050/comp/renderers
Created folder: SH050/cache
Created folder: SH050/cache/fx
Created folder: SH050/cache/anim
Created folder: SH050/publish
Created folder: SH050/publish/images
Created folder: SH050/publish/videos
Shot SH050 created successfully!
```

What You Must Submit

- ✓ Your script (shotsetup.py)
- ✓ Your JSON file (shot_structure.json)
- ✓ Screenshot of terminal output
- ✓ Screenshot of generated folder structure in file explorer